Transfer Learning

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State

Professor, Electrical and Computer (ECE) Engineering, NC State

NAIRR AI Education Fellow, Computing Research Association (CRA)

Why Train From Scratch Is Expensive

The data-hungry, compute-intensive costs from earlier, made concrete

TRAINING FROM SCRATCH

A model like ResNet-50 was trained on roughly 1.2 million labeled images, using days of multi-GPU compute — before it could recognize anything useful.

YOUR ACTUAL PROBLEM

A few hundred labeled weld-defect images, and a training budget measured in hours on a single GPU, not days on a cluster.

That gap is exactly what transfer learning is designed to close.

The Core Idea: Reuse, Don’t Restart

Start from a model that already learned general-purpose patterns

PRE-TRAINED MODEL

Trained on a large, general dataset (ImageNet, a large text corpus)

→

REUSED & ADAPTED

General features carried over, task-specific parts adjusted

→

YOUR TASK

A small, specific dataset — weld defects, your own documents

“Pre-trained” just means someone already paid the data-and-compute cost — you’re borrowing the result.

Why It Works: Not All Layers Are Equally Task-Specific

A direct callback to the depth-and-hierarchy idea from the CNN session

Reusable across almost any task → → → Needs adaptation for your specific task

EDGES
GENERAL

→

TEXTURES
GENERAL

→

PARTS
MOSTLY GENERAL

→

OBJECT
TASK-SPECIFIC

This is why you can freeze the early layers and only retrain the last one or two — they were never task-specific to begin with.

Two Strategies: Feature Extraction vs. Fine-Tuning

Different amounts of the pretrained model get updated

Feature Extraction

Freeze the pretrained layers entirely; only train a small new “head” on top for your specific task.

Best when: your dataset is small and similar to the pretraining data.

Fine-Tuning

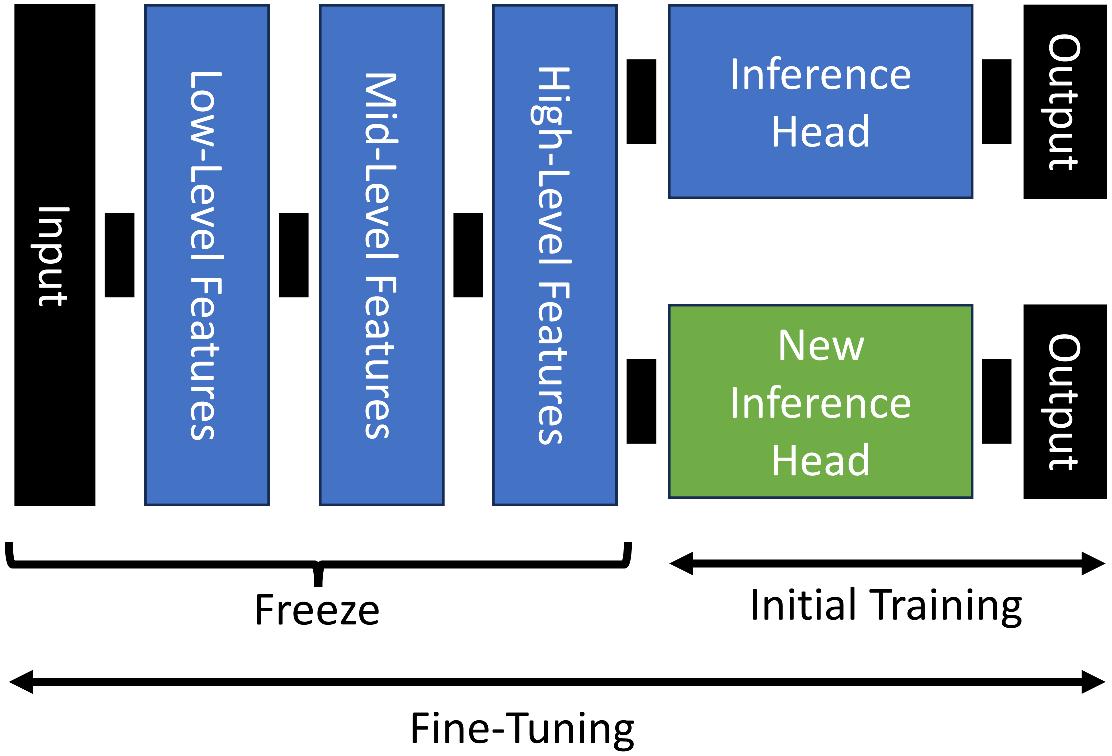
Unfreeze some or all pretrained layers and keep training them, usually at a lower learning rate.

Best when: you have more labeled data, or your task differs more from the pretraining data.

Both start from the same pretrained checkpoint — the only difference is how much of it you let training touch.

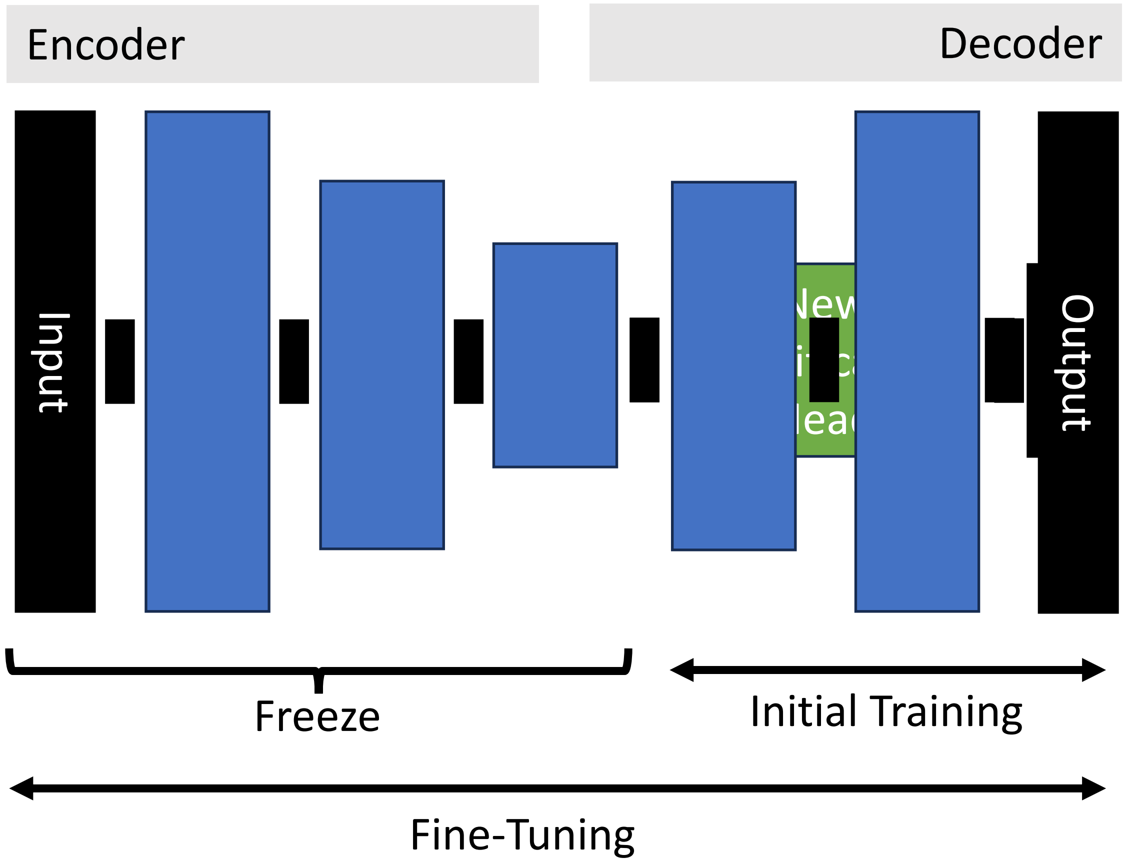
Transfer Learning: Across Datasets

- Across datasets of the same format, we can **pre-trained** model by removing its head and replacing it with a new head.
- The new head is trained by **freezing** the rest of the network.
- This is often followed by **fine-tuning** of the entire network.



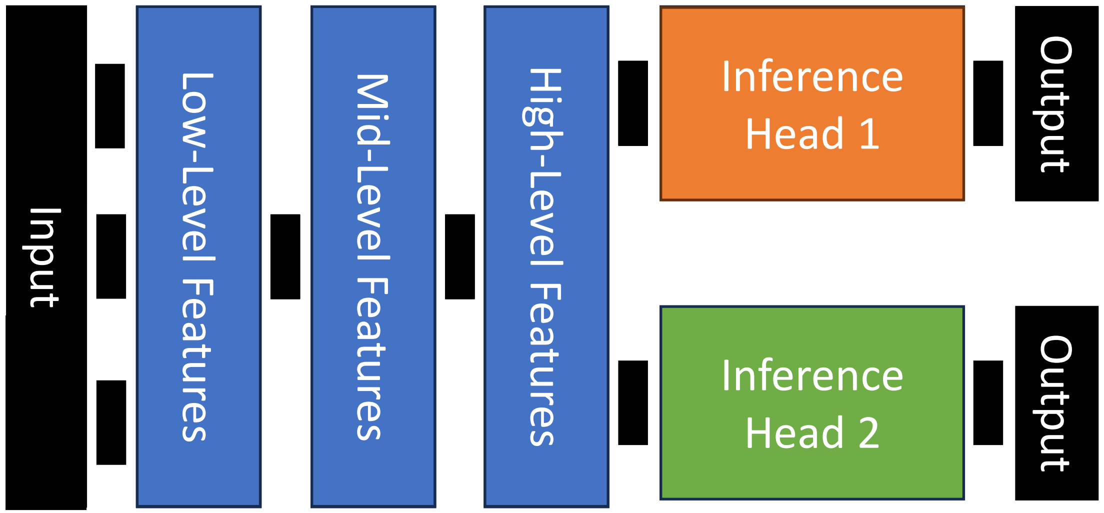
Transfer Learning: Semi Supervised

- Autoencoder are an option when there is no labeled data available in the same format from other datasets, but there is enough unlabeled data.
- We use the encoder portion of the network and attach an inference head.



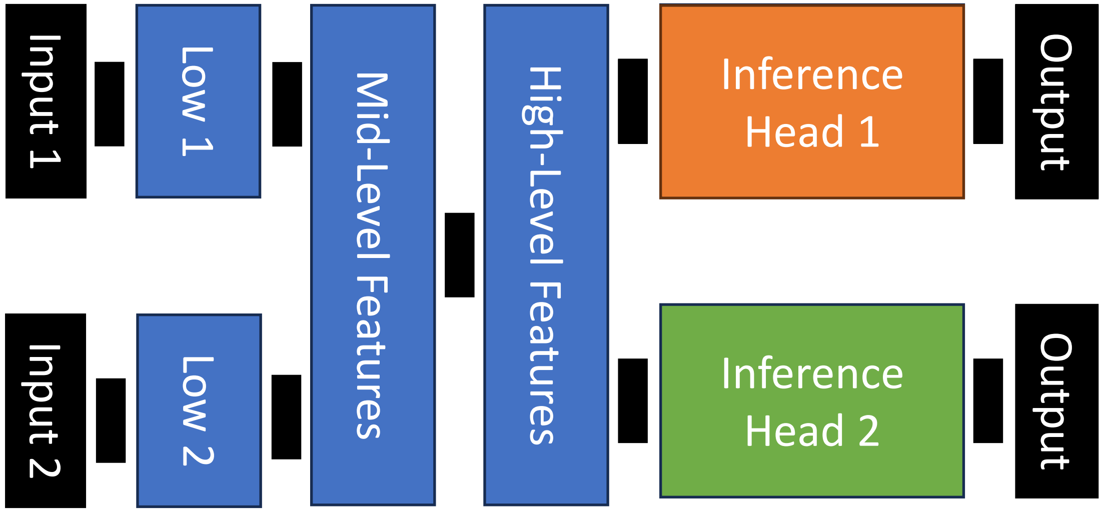
Multi-Task Learning

- It can be used to train multiple inference tasks jointly so they can help each other.
- Useful when one or all the datasets are relatively small.
- The cost is the sum of the cost for each inference task separately applied to their corresponding dataset. During training we use a mixed batch.



Multi-Task Learning

- Some variants may incorporate separate learning of low-level features.
- Useful when considering images of different modalities for similar tasks, e.g., using infrared images for detection of people, and color images for general object detection.



Choosing a Strategy: Data × Task Similarity

NC STATE Engineering

A simple decision framework for which approach to reach for

← Data Size →

↓ Task Similarity ↑

Small Data + Similar Task

Feature extraction — freeze almost everything, train a small new head.

Large Data + Similar Task

Fine-tune the later layers, or the whole network at a low learning rate.

Small Data + Different Task

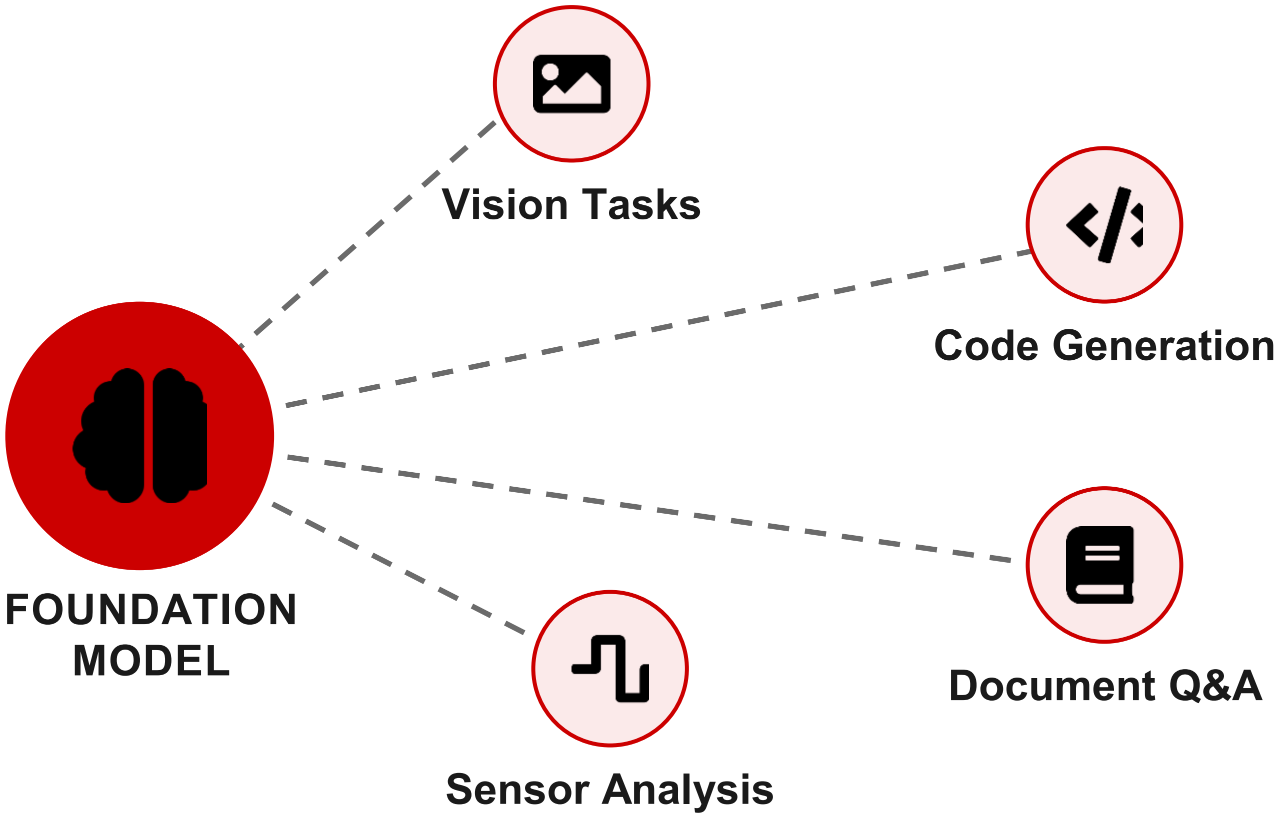
Trickiest case — feature extraction from earlier layers or find a closer pretrained model.

Large Data + Different Task

Fine-tune extensively or consider training from scratch after all.

Foundation Models: The Current Extreme

One massive pretrained model, adapted across countless downstream tasks



Same underlying idea as before, taken to the extreme: one very expensive pretraining run, reused millions of times over.

Limitations & Watch-Outs

Transfer learning isn't automatic — and it isn't free

Domain Mismatch

A model pretrained on natural photos may not transfer well to thermal or X-ray imagery — the features it learned may not apply.

Catastrophic Forgetting

Fine-tuning too aggressively, or on too little data, can erase the general knowledge you were trying to reuse.

Licensing & Compute

Large foundation models still carry real compute and licensing costs for fine-tuning and deployment.

Transfer Learning

https://forms.gle/X5ncinR3TZqLp7eQA